

Figure 1. D1 sub-score → grade reference matrix (DO/ORP-only main link, final)

criterion		A (≥4.5)	B (3.5–4.5)	C (2.5–3.5)	D (1.5–2.5)	F (<1.5)
Q_spike (spike)	spike rate ≤ 2%	very rare	rare	occasional	frequent	very frequent (>20%)
Q_step (step)	KS statistic	near 0	low	moderate	elevated	high (sustained)
Q_drift (drift)	PLS residual z	z < 1.5	1.5–2.0	2.0–2.5	2.5–3.0	> 3.0 sustained
Q_freeze (freeze)	RLE / unique ratio	no freeze	<5 min	5–15 min	15–60 min	>60 min
Q_regime (regime)	W1 normalised	< 1.0	1.0–2.0	2.0–3.0	3.0–4.0	> 4.0